

Roshan Balu T M B

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WORK EXPERIENCE

Automation System Engineer

Jan 2026 - Present

Sims Lifecycle Services, West Chicago, Illinois, USA

- **Working with external automation vendors and cross-functional teams** to build robotic workcells, driving integration of MES platforms, OEE analytics, and quality testing metrics.
- **Managed and mentored a software engineering team**, guiding the agile development and continuous maintenance of enterprise-grade Machine Vision solutions.

Mechatronics Engineer

Sep 2023 - Jan 2026

Sims Lifecycle Services, West Chicago, Illinois, USA

- **Led development and integration of Industrial Machine Vision systems** for Quality Assurance, utilizing GigE cameras and Keyence sensors for high-speed barcode scanning and OCR.
- **Deployed Mitsubishi and Fanuc robotic systems** for precise electronic component feeding operations, optimizing production line efficiency.
- **Developed custom Computer Vision models (YOLOv8)** for automated component identification and inspection, achieving high accuracy in QA processes.
- **Integrated vision/robotic systems with MES** using Python and REST API for seamless data exchange and process control via **Azure DevOps CI/CD pipelines** for Windows installation, managing all associated CAD assets within **Autodesk Vault PDM**
- **Managed full project lifecycle** of external vendor automation projects, including developing detailed SOPs and validating FAT/SAT protocols.
- **Increased production throughput by 30%** via automated machine vision systems, simultaneously **reducing operator strain and training efforts by 80%**.

Graduate Researcher - MAVs

Apr 2022 - Aug 2023

Agile Robotics and Perception Lab, NYU, NYC, USA

- Created a prototype for autonomous charging of UAVs using electromagnetic contacts and a programmed routine. Modified a balance charger board and designed a custom enclosure and PCB for the electrical contacts. **Reduced Human intervention by 100%**
- Design, simulation and fabrication of mechanical and electro-mechanical components for Aerial Robots & creating documentation for hardware and software, maintaining BOM (PLM), managing 3D printers (FDM&SLA)

Mechanical Engineer - Industrial Automation

Sep 2018 - Jul 2021

SEP India Pvt Ltd, Hosur, India

- Developed a **SCADA** system by integrating internally designed autoloader robots and a custom-developed **vision system**, similar to Keyence/Cognex. Enabled real-time process control, inspection, and correction, while providing online reporting (AWS Dashboard) for live updates (IoT).
- Spearheaded the design and development of a fully autonomous manufacturing cell with Autoloaders, special jigs and fixtures with an auto-inspection line, enabling live updates of production status and machined part quality. This resulting in a significant **35% increase** in production compared to human-operated production.

EDUCATION

Master of Science in Mechatronics and Robotics

Sept 2021 - May 2023

New York University (NYU)

Coursework: Foundation of Robotics (TA), Advanced Mechatronics, Reinforcement Learning & control

Bachelor of Technology in Mechanical Engineering

July 2014 - May 2018

Amrita School of Engineering, Coimbatore, India

Passed with First Class & Distinction

PUBLICATIONS

- Jennifer Yeom, **Roshan Balu T M B**, Guanrui Li, Giuseppe Loianno, **Experimental System Design of an Active Fault-Tolerant Quadrotor**, ICUAS 2024 [Publication](#)
- Alessandro Saviolo*, Jeffrey Mao*, **Roshan Balu TMB***, Vivek Radhakrishnan, Giuseppe Loianno, **AutoCharge: Autonomous Charging for Perpetual Quadrotor Missions**, ICRA 2023. [Publication](#) [Video](#)
- P.Prudhvi Sai*, **T.M.B.Roshan Balu***, R.Vaira Vignesh, Chinta Vikram, Bhaskara Sastry, R.Padmanaban, **Artificial neural network models for predicting the corrosion behavior of friction stir processed AA5083**, Materials Today: Proceedings, 2021, ISSN 2214-7853 [Publication](#)

CERTIFICATIONS AND TRAINING

- **2025** | Advanced Robot Training | Mitsubishi Electric | Vernon Hills, IL
- **2025** | Machine Vision Training | Keyence Corporation | Online
- **2024** | CV for Inspection & Fundamentals of CUDA | Nvidia GTC | Online

SKILLS

Programming

Python, C++, Matlab, Bash, Git, PLC

Robotics

Robot Operating System (ROS), ROS2, RT Toolbox3, Roboguide, TP programming

CAD/CAE Softwares

Autodesk Inventor (with Vault), Solidworks (with PDM), Eagle (PCB), NX Siemens

Prototyping

3D printing (FFF/FDM & SLA), Laser cutting, CNC Machining